Increase of temperature and decrease of oxygen concentration give an early indication of a decomposition.

In addition, should decomposition occur, the residue left after decomposition may have only half the mass of the original cargo. Due consideration shall be given to the effect of the loss of mass on the stability of the ship.

Discharge
Organic contamination aggravates decomposition behaviour in the presence of sources of heat, and therefore bunkering of fuel oil shall not be allowed during discharge.
Pumping of fuel oil in spaces adjacent to the cargo spaces for this cargo, other than the engine-room, shall not be allowed during discharge.

SOURCE: MSC.500(105).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf
RESOLUTION MSC.500(105) (adopted on 28 April 2022)
AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
- 13 -

This cargo is hygroscopic and may cake in overhangs, impairing safety during discharge.

If this cargo has hardened, it shall be trimmed to avoid the formation of overhangs, as necessary.

Clean-up
After discharge of this cargo, the bilge wells and the scuppers of the cargo spaces shall be checked, and any blockage in the bilge wells and the scuppers shall be removed.

Emergency procedures
Special emergency equipment to be carried
Protective clothing (boots, gloves, coveralls and headgear)
Self-contained breathing apparatus

Emergency procedures
Wear protective clothing and self-contained breathing apparatus

Emergency action in the event of fire or decomposition
Decomposition in a cargo space containing this material : The master and officers are to
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SOURCE: MSC.500(105).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf
Emergency action in the event of fire or decomposition
Decomposition in a cargo space containing this material : The master and officers are to
note that the ship's fixed gas fire -fighting installation will b e ineffective on decompositions involving this cargo and must not be used. If decomposition is identified, water must be applied

without delay. Injection to the seat of decomposition is the first control measure (e.g. using Victor lance) because it uses le ss water and can be more effective in early decomposition stages. Total flooding is the final control measure but can introduce stability and stress issues. The consequential risk to the stability of the ship through fluidization of the cargo must be taken into account in both cases. Application of water to the surface of the cargo is much less effective and can give a false sense of safety.
Fire in an adjacent cargo space : Heat transferred from fire in an adjacent space can cause
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SOURCE: MSC.500(105).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf
and can give a false sense of safety.
Fire in an adjacent cargo space : Heat transferred from fire in an adjacent space can cause the material to decompose with consequent evolution of toxic fumes. Open hatches to provide maximum ventilation. Dividing bulkheads should be cooled.

Medical first aid
Refer to the Medical First Aid Guide (MFAG), as amended

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"CLAM SHELL

This schedule shall only apply to whole clam shells.

Description
This cargo is a by -product generated in the process of clam farming. Dark grey to beige, granular in form, not soluble, solid and natural material.

Characteristics
Physical properties
Size Angle of repose Bulk density (kg/m3) Stowage factor (m3/t)
5 mm to 30 mm 34° 1,058 0.945
Hazard classification
Class Subsidiary hazard(s) MHB Group
Not applicable Not applicable Not applicable C
RESOLUTION MSC.500(105) (adopted on 28 April 2022)
AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
- 14 -

SOURCE: MSC.500(105).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf
RESOLUTION MSC.500(105) (adopted on 28 April 2022)
AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE
- 14 -

Hazard
No special hazards.
This cargo is non-combustible or has a low fire risk.

Stowage and segregation
No special requirements.

Hold cleanliness
No special requirements.

Weather precautions
No special requirements.

Loading
Trim in accordance with the relevant provisions required under sections 4 and 5 of this Code.

Precautions
No special requirements.

Ventilation
No special requirements.

Carriage
No special requirements.

Discharge
No special requirements.

Clean-up
After discharge of this cargo, the cargo spaces and the bilge wells shall be swept clean and then thoroughly washed out."

"LEACH RESIDUE CONTAINING LEAD

Description
Intermediate by-product formed as a result of the hydrometallurgical production of zinc and/or
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SOURCE: MSC.500(105).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.500(105).pdf
"LEACH RESIDUE CONTAINING LEAD

Description
Intermediate by-product formed as a result of the hydrometallurgical production of zinc and/or zinc compounds. Insoluble grey to brown granular substance obtained during dissolution of zinc ores or concentrate in sulphuric acid to produce zinc sulphate solutions after physical separation such as flotation and filtration.

Characteristics
Physical properties
Size Angle of repose Bulk density (kg/m3) Stowage factor (m3/t)
Fines up to 100 µm 45° to 52° 800 to 1,600 0.63 to 1.25
Hazard classification
Class Subsidiary hazards MHB Group
Not applicable TX and CR A and B

Hazard
This cargo may liquefy if shipped at a moisture content in excess of its transportable moisture limit (TML). See sections 7 and 8 of this Code.

Harmful if swallowed or inhaled.
RESOLUTION MSC.500(105) (adopted on 28 April 2022)
AMENDMENTS TO THE INTERNATIONAL MARITIME SOLID BULK CARGOES (IMSBC) CODE

SOURCE: MSC.502(105).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.502(105).pdf
RESOLUTION MSC.502(105) (adopted on 28 April 2022)
AMENDMENTS TO THE CODE OF SAFETY FOR SPECIAL PURPOSE SHIPS, 1983
(1983 SPS CODE)
MSC 105/20/Add.1
Annex 9, page 1
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ANNEX 9
RESOLUTION MSC.502(105)
(adopted on 28 April 2022)
AMENDMENTS TO THE CODE OF SAFETY FOR SPECIAL PURPOSE SHIPS, 1983
(1983 SPS CODE)
THE MARITIME SAFETY COMMITTEE,
RECALLING Article 28(b) of the Convention on the International Maritime Organization concerning the functions of the Committee,
RECALLING ALSO resolution A.534(13), by which the Assembly, at its thirteenth session, adopted the Code of Safety for Special Purpose Ships, 1983 (1983 SPS Code),
RECALLING FURTHER that the Assembly authorized the Committee to amend the Code as may be necessary,
NOTING the amendments to the International Convention for the Safety of Life at Sea, 1974 ("the Convention") adopted by resolution MSC.496(105),
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SOURCE: MSC.502(105).pdf
URL:
https://wwwcdn.imo.org/localresources/en/KnowledgeCentre/IndexofIMOResolutions/MSCResolutions/MSC.502(105).pdf
NOTING the amendments to the International Convention for the Safety of Life at Sea, 1974 ("the Convention") adopted by resolution MSC.496(105),
HAVING CONSIDERED, at its 105th session, consequential amendments to the Special